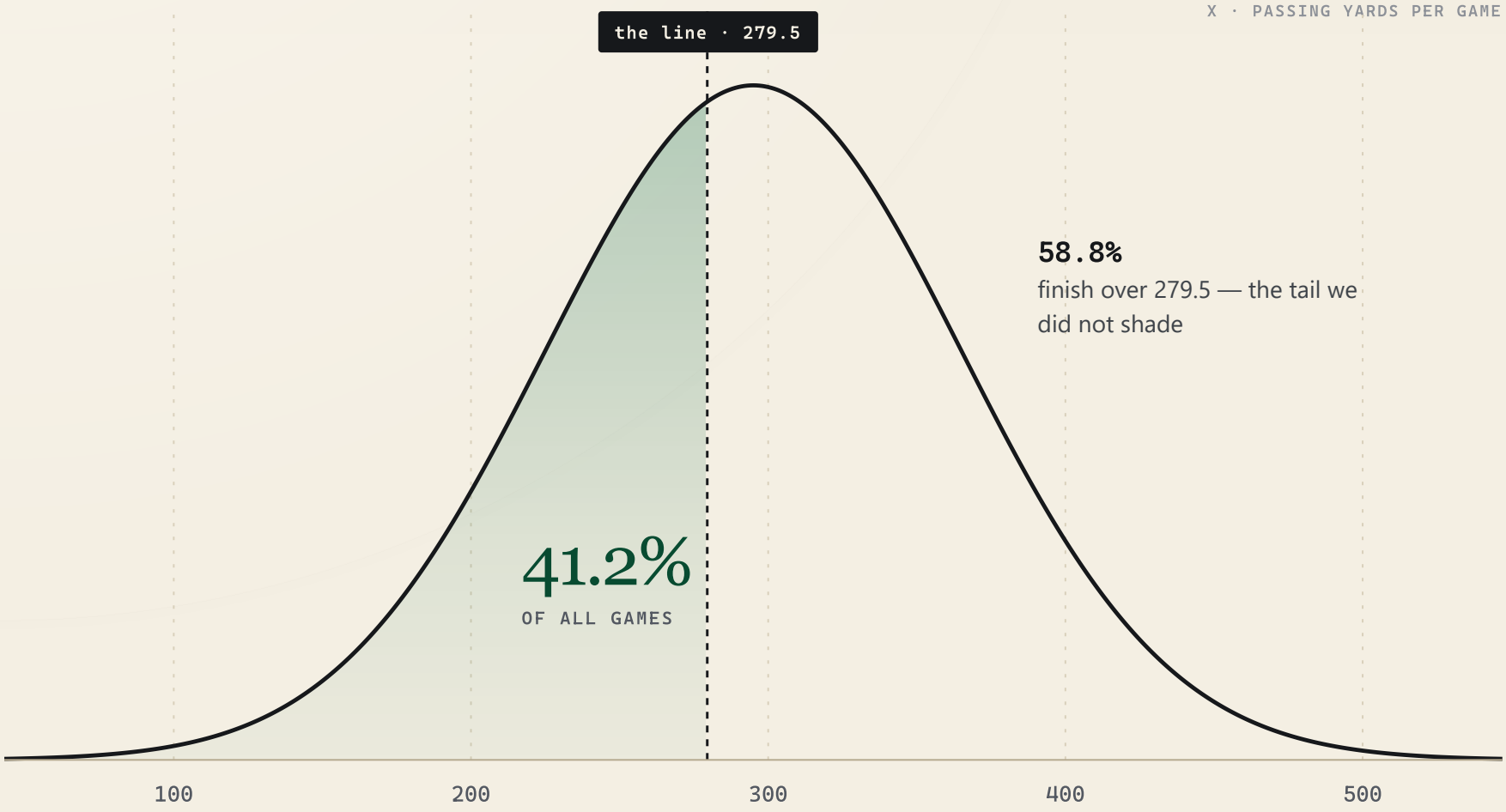
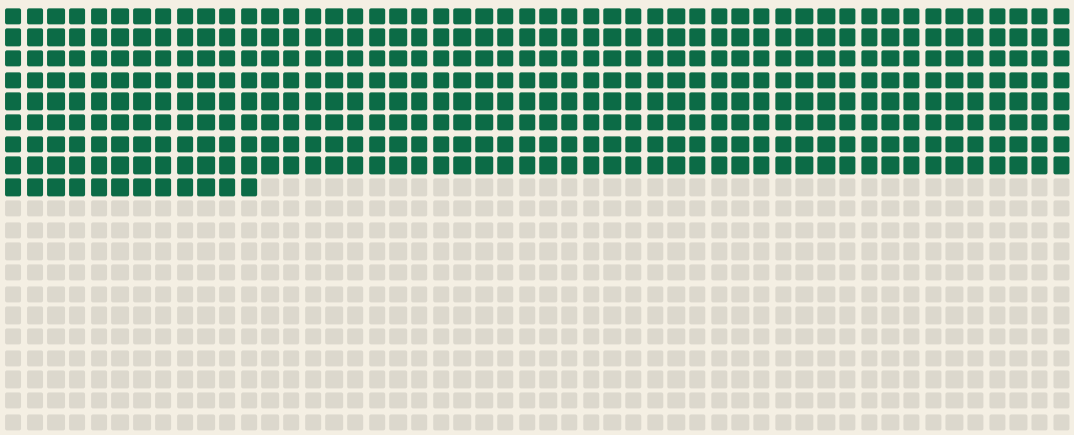
An area is a count.

The shaded region under a curve is not a shape — it is a **fraction of games**. Count the outcomes inside the region, divide by every outcome the model allows, and you have read a cumulative distribution function.





412 UNDER THE LINE 588 OVER

One cell, one game. The **expected tally** across 1,000 games: 412 land under 279.5, 588 land over. The area is that fraction — 412 ÷ 1,000.

0.412	=	412	÷	1,000	=	41.2%
Φ(Z) – AREA LEFT OF THE LINE		GAMES UNDER 279.5, EXPECTED		EVERY GAME THE MODEL ALLOWS		THE AREA, READ AS A SHARE

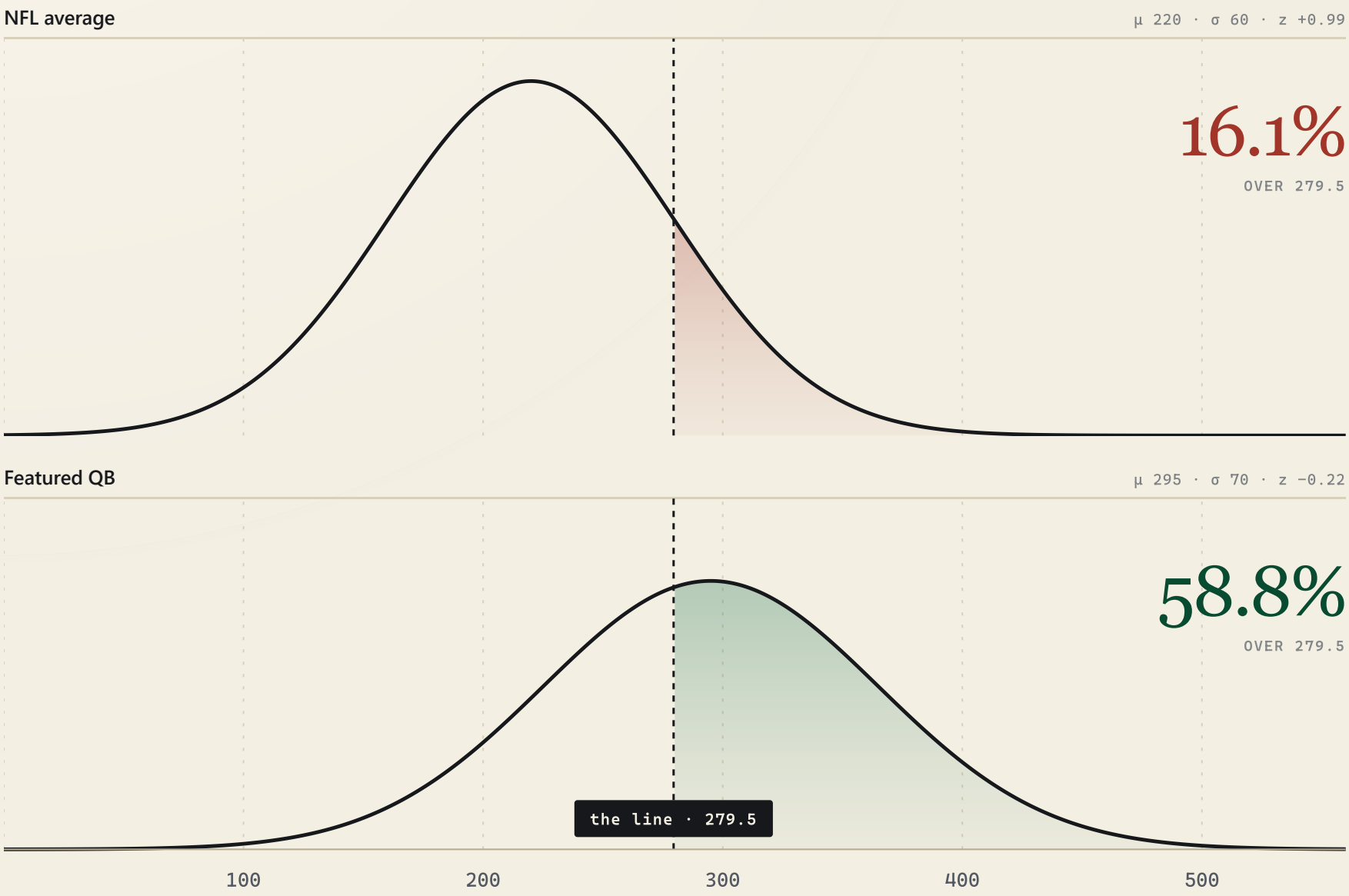
> A probability is a fraction with a very large denominator. The CDF is just that fraction, drawn for you — z = (279.5 − 295) / 70 = −0.221.

POSTER 1 OF 6 · THE CDF AS A COUNTER

Model: passing yards per game, normal(μ 295, σ 70); prop line 279.5. Every value from `scripts/model.py`. A 400,000-draw Monte Carlo lands within 0.12 points of the analytic area.

The line is the same. The area isn't.

A line is not a probability — it is a **question**. Ask it of the NFL average and you get a sliver of the curve. Ask it of one starting quarterback and you get most of it. The number is identical. The **z-score**, and therefore the area, is not.



Same shape, stretched differently. The average offense is tight around 220 yards; the featured QB sits 75 yards higher with a wider spread. Standardize each curve and the single line 279.5 turns into two different z-scores — and two different areas.

	← UNDER THE LINE	OVER THE LINE →	COUNT / 1,000
NFL average μ 220 · σ 60	83.9% under	16.1% over	161 / 1,000
Featured QB μ 295 · σ 70	41.2% under	58.8% over	588 / 1,000

SAME CUT

279.5

one threshold, two questions

CURVE CENTERS

220 → 295

the two curve centers sit 75 yards apart

TAIL RATIO

3.66×

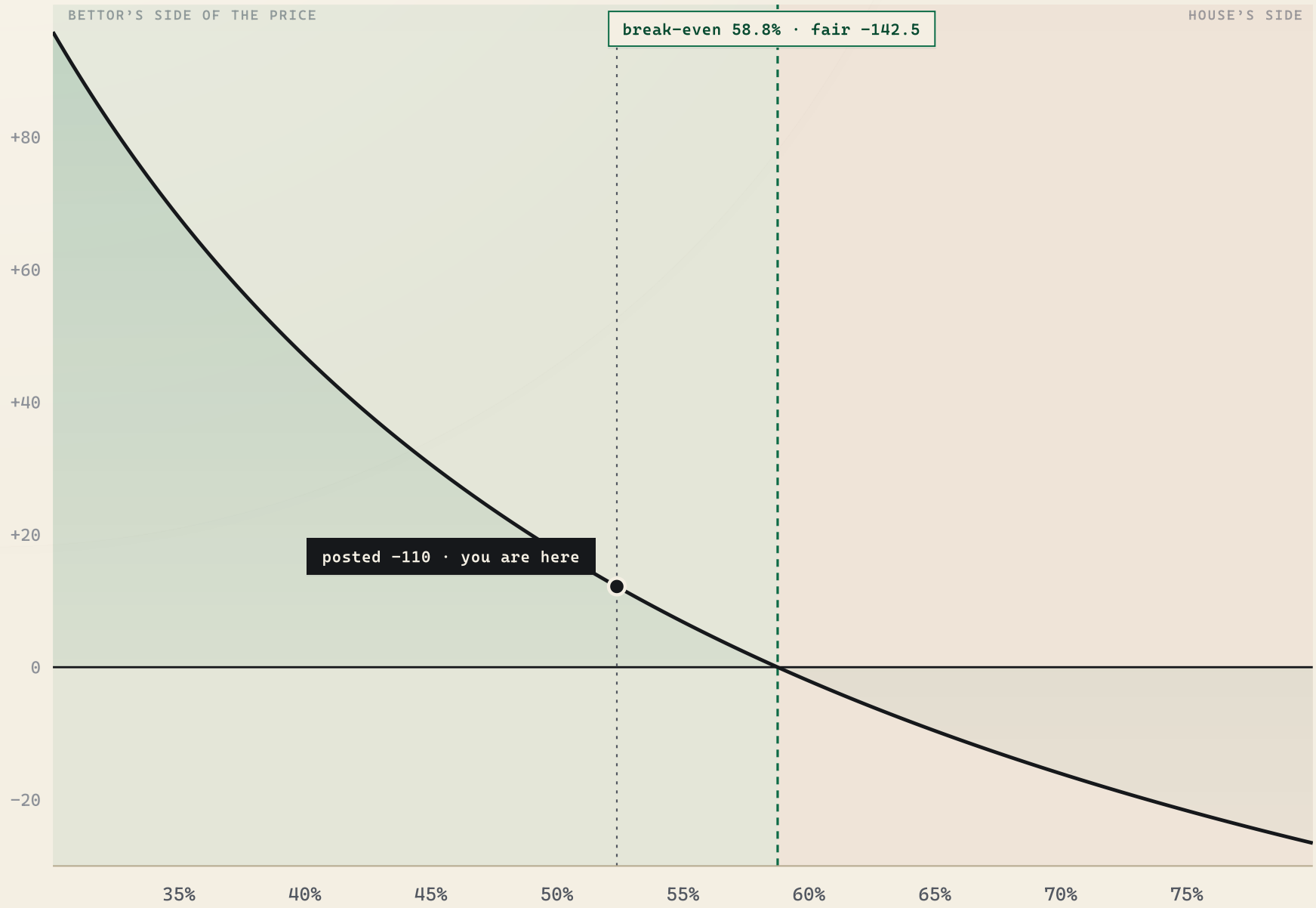
how much likelier the QB is to clear it

> Books don’t move the line because the arithmetic is wrong. They move it because the curve underneath the number changed.

League normal(μ 220, σ 60) at 279.5 → z = +0.99, P(over) = 16.1%. QB normal(μ 295, σ 70) at 279.5 → z = −0.22, P(over) = 58.8%. The QB tail is 3.66× the league tail. Both panels share one x-scale and one y-scale.

A price is a probability too.

−110 doesn’t mean 50-50. It means **52.4%**, because the vig rides inside the price. The CDF says this QB clears 279.5 **58.8%** of the time, which makes the fair price −142.5. The 6.4-point gap is the entire ticket.



x: the price, read as an implied probability. y: expected profit on a $100 wager. Left of the dashed green line the payout is better than the CDF says it should be; right of it, you are paying a tax. The posted −110 sits on the paying side.

MARKET SAYS

52.4%

−110 implies 110 ÷ 210, vig included

THE CDF SAYS

58.8%

P(over 279.5) for this QB

FAIR PRICE

−142.5

where the curve breaks even

EXPECTED VALUE PER $100 WAGERED

+$12.18

58.76% × $90.91 − 41.24% × $100 = **+$12.18**

edge 6.4 points · ROI 12.2%

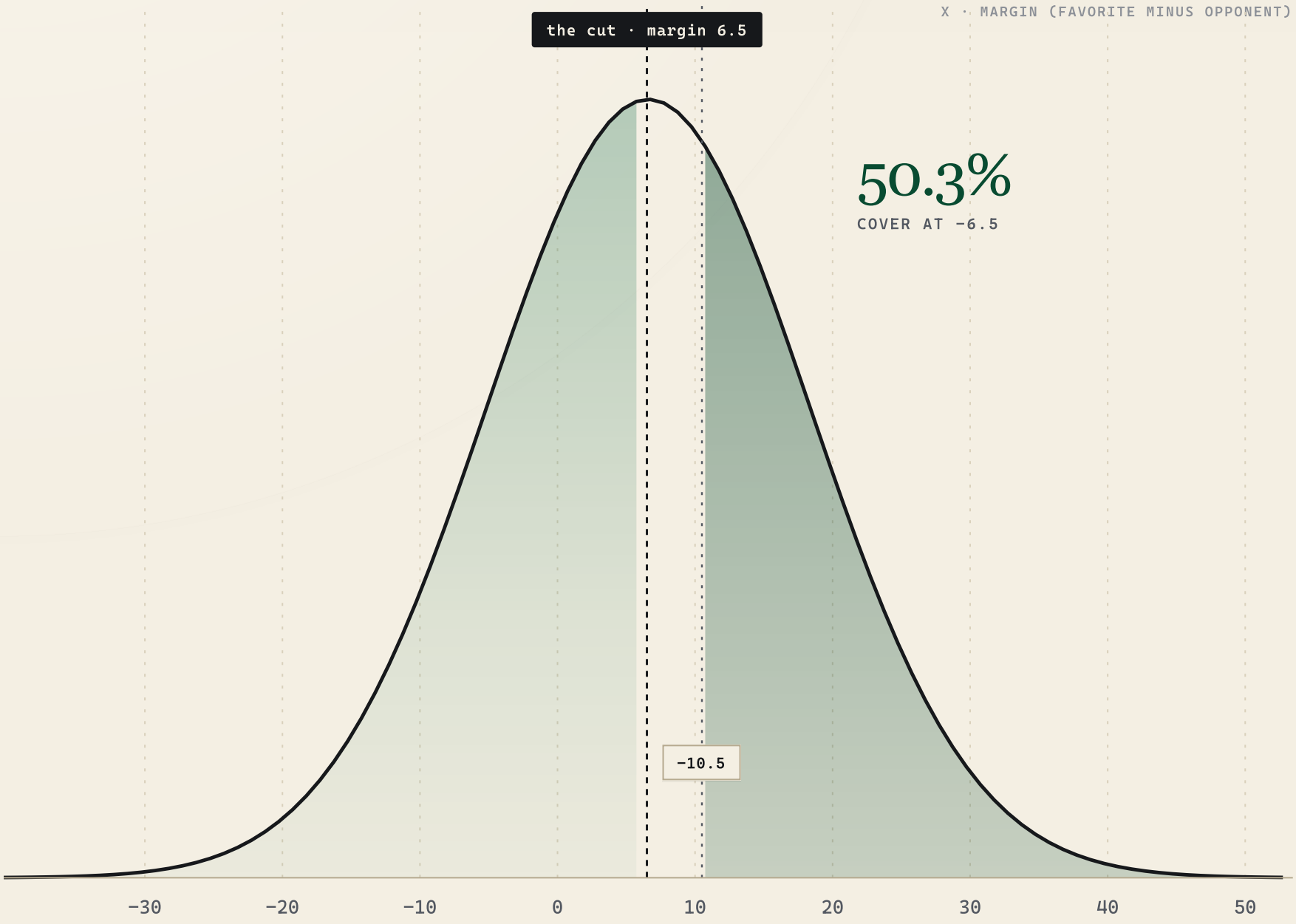
> The CDF never tells you who wins. It tells you what the ticket is worth — and a book that leaves a 58.8% shot at −110 has already told you where the line is going.

POSTER 3 OF 6 · PRICE, PROBABILITY, AND THE VIG

EV per $100 wagered = p(1 − q)/q − (1 − p), where q is the price’s implied probability. The curve crosses zero exactly where q = p = Φ((279.5 − 295) / 70) = 58.76%. The other side of the same market: under 279.5 at −110 is −$21.27 per $100.

A spread is a tail, not a range.

A spread isn't a band of final scores — "wins by three to seven" is a different question. It's one number, subtracted once, so **covers** is a single cut across the margin curve. The book puts that cut where the area is half on each side, then charges both sides for it.



The tail right of the cut is "covers −6.5" — 50.3% of the margin curve. The lighter wedge between the two lines is what a book takes away by moving the number four points to −10.5: cover falls to 37.0%, 13.4 points of area gone. On a whole-number line the margin can land exactly on the number, and no shaded area can represent that third outcome.

$1 - \Phi(z)$ THE TAIL PAST THE CUT = **1 − Φ(−0.01)** Z = (6.5 − 6.60) / 11.72 = **50.3%** THE AREA THAT COVERS AT −6.5

THE CUT	COVER AT 6.5	MOVE TO 10.5
−6.5	**50.3%**	**37.0%**
one threshold on the margin, not a range of scores	the area right of the cut — half, by construction	four points of line = 13.4 points of area

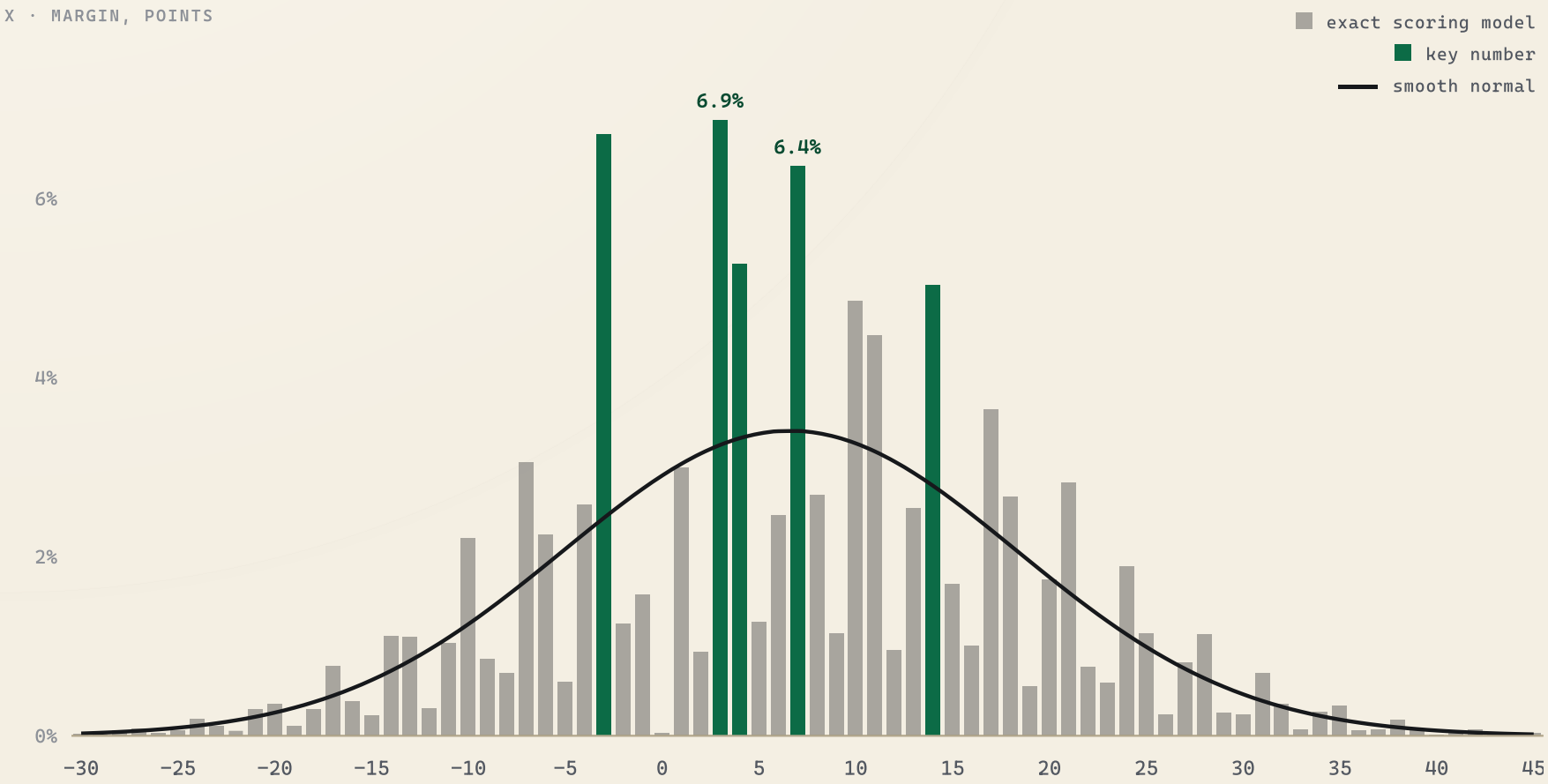
> The spread is not a prediction of the score. It is the cut where the two tails balance — and every point the book moves is paid for in area.

POSTER 4 OF 7 · A SPREAD IS ONE THRESHOLD ON A MARGIN

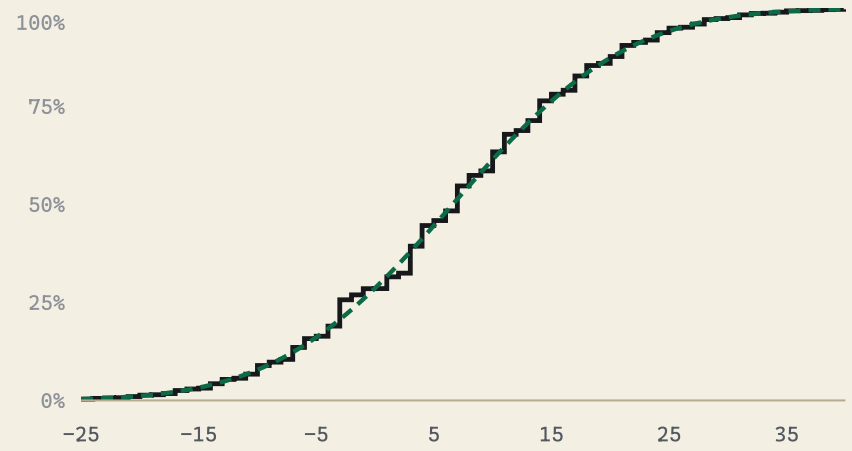
Margin = favorite score − opponent score, modeled normal(μ 6.60, σ 11.72); μ and σ are read off the discrete scoring model on 03.5, so the smooth picture and the lumpy one agree. At the half-point line, P(cover) = 50.3%. A whole-number line cannot be priced by the curve: at −7 the margin lands exactly on the number in 6.4% of games.

The curve is blind at 3 and 7.

Football scores arrive in 3s and 7s, so the margin distribution grows lumps at exactly those numbers. A normal curve spreads the same mass evenly: it gives 7 a **3.4%** slice, while the scoring model puts **6.4%** there. On a whole-number spread, that lump is a push — a third outcome with no area.



THE CDF, UN-SMOOTHED — each jump is a point mass



A discrete CDF is a staircase. Each vertical jump is the probability of an exact margin — the same numbers the bars carry. Φ is the diagonal it is trying to be.

KEY NUMBERS — probability of landing exactly there

MARGIN	SCORING MODEL	SMOOTH CURVE	RATIO
+3	6.9%	3.2%	2.1×
−3	6.7%	2.4%	2.8×
+7	6.4%	3.4%	1.9×
+4	5.3%	3.3%	1.6×
+14	5.0%	2.8%	1.8×

The lumps are the scoring grid: 3 = field goal, 7 = touchdown, and 10 or 14 are combinations. The smooth model averages that structure away.

P(MARGIN = 3)

6.9%

the smooth curve says 3.2%

P(MARGIN = 7)

6.4%

the smooth curve says 3.4%

COVER AT −6.5

51.7% vs 50.3%

the lumps move the answer 1.3 points

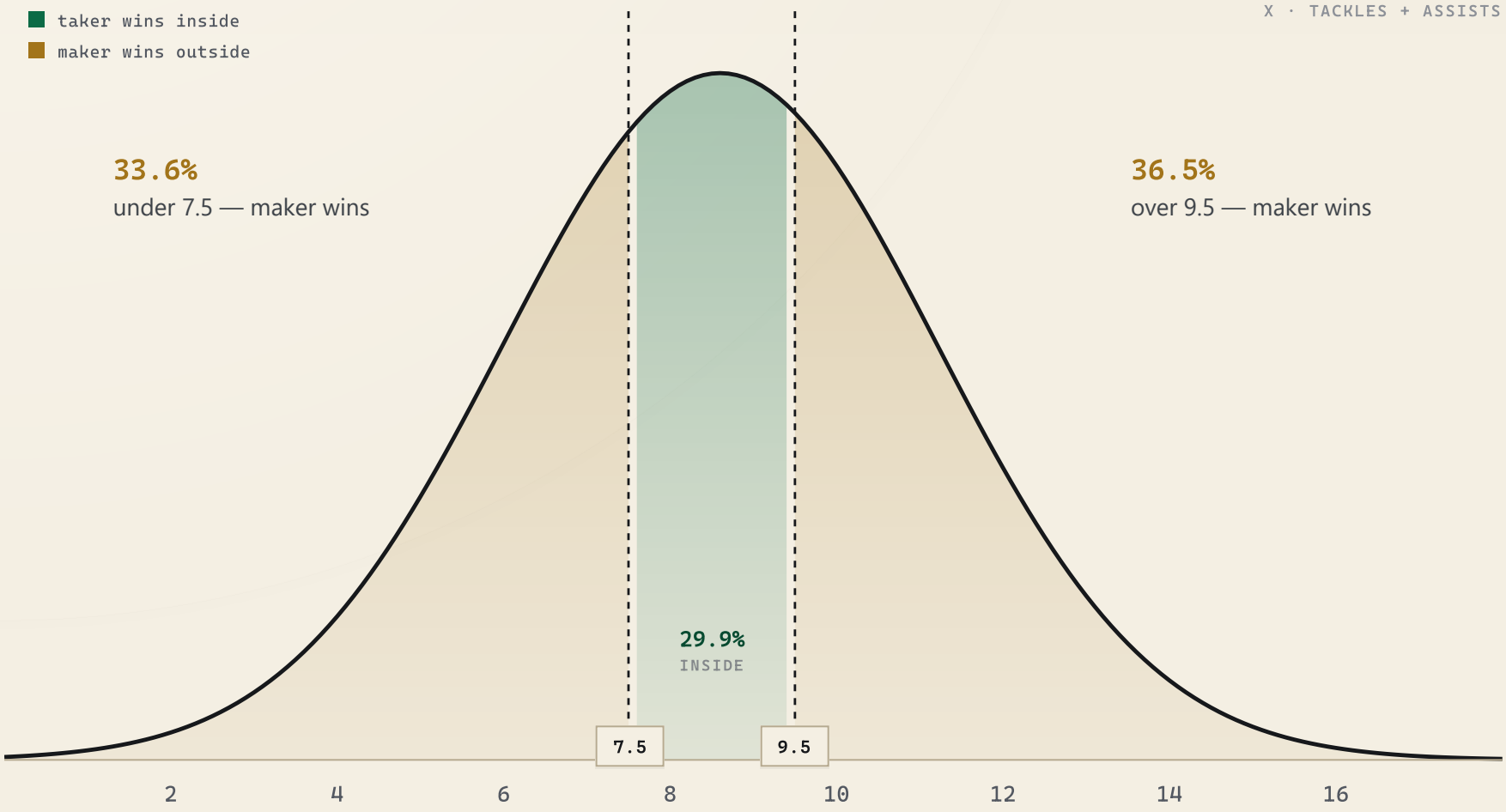
> A half-point isn’t there to make the bet fair. It’s there to delete the bar.

POSTER 5 OF 7 · WHEN THE OUTCOME MATTERS, THE PMF RULES

Model: 11 drives per team, each ending in nothing, a field goal (3) or a touchdown plus XP (7); ties resolved by overtime. Exact convolution, no simulation. The normal shares the model’s μ 6.60 and σ 11.72, so the comparison is like for like. The axis is clipped at ±30 points; the omitted tails carry under 1% of games.

A range is a subtraction.

Two lines on one player split the curve three ways: below, inside, above. “Inside” is the only region that isn’t a tail — it is one CDF read minus another, Φ(9.5) − Φ(7.5). With no house and no vig, the maker’s side is the exact mirror of the taker’s.



OUTCOME	PROBABILITY	TAKER, PER UNIT	MAKER, PER UNIT
under 7.5 (≤ 7)	33.6%	−0.426	+0.426
inside 8–9	29.9%	+1.080	−1.080
over 9.5 (≥ 10)	36.5%	−0.494	+0.494
expected value	100.0%	+0.000	+0.000

P(INSIDE)

29.9%

Φ(9.5) − Φ(7.5), the area between two cuts

FAIR UNITS, EACH LEG

1.506 / 1.574

over 7.5 / under 9.5, no vig

HOUSE EDGE

0.000

both sides’ EV is exactly zero — the market is the whole book

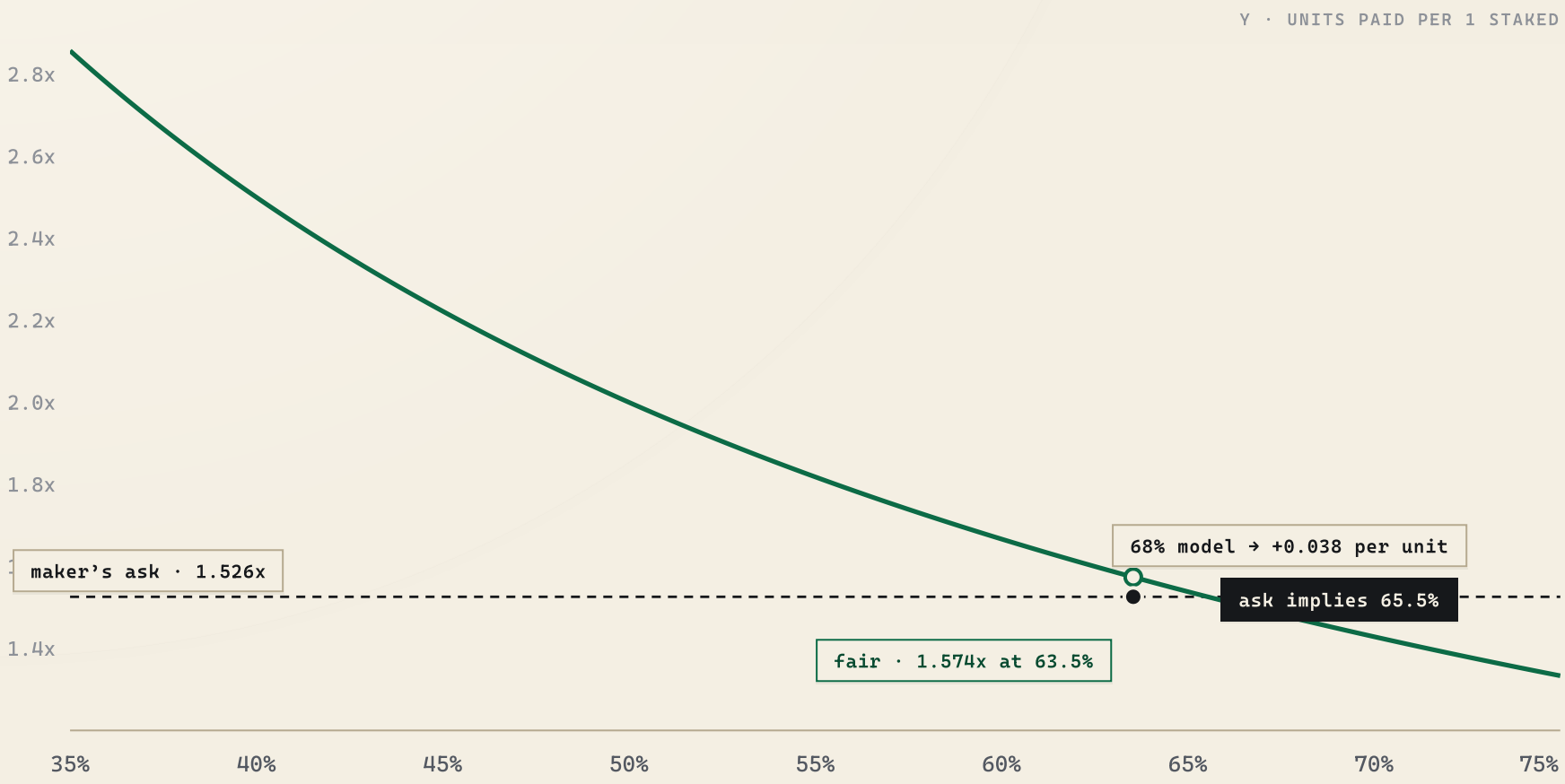
> A middle isn’t a loophole. It’s the price of the inside — one side buys the range, the other sells it, and the two payoffs add to zero in every outcome.

POSTER 6 OF 8 · THE RANGE IS Φ(B) − Φ(A)

Player: tackles + assists, normal(μ 8.6, σ 2.6); lines 7.5 and 9.5, so the integer outcomes 8 and 9 are the middle. Fair units = 1 / probability with no vig, and payoffs are per 1 unit staked on each leg. Φ(0.346) − Φ(−0.423) = 29.93%.

Same division, both directions.

A maker starts with a percentage and needs units. A taker sees units and needs a percentage. Both are **1/p** — which is why a price is a probability in payout clothing, and why checking a bet runs both ways.



THE MAKER

Starts from his own number: **63.5%** for under 9.5. He wants **2.0 points** of edge, so he asks a price that implies 65.5%: units = 1 / 65.5% = **1.526x**. If a taker accepts, he is paid 0.031 units per unit staked — his edge, in units.

THE TAKER

Reads the same 1.526x backwards: 1 / 1.526 = 65.5% implied. His own model says 63.5%, so his EV is **−0.031** per unit and he passes. A taker whose model says 68% gets **+0.038** per unit and takes it. Same price, different answer.

THE CONVERSION TABLE

the same number, three costumes

PROBABILITY	UNITS	AMERICAN
50.0%	**2.000x**	−100
55.0%	**1.818x**	−122
60.0%	**1.667x**	−150
63.5%	**1.574x**	−174
65.5%	**1.526x**	−190
70.0%	**1.429x**	−233

MAKER'S ASK

1.526x

implies 65.5% — two points over his own model

MAKER'S EDGE

+0.031

units per unit staked, if a taker agrees

TAKER'S CHECK

−0.031 / +0.038

at 63.5% vs at a 68% model — same price

> No vig doesn't mean no edge. It means one side's edge is exactly the other side's price — and the only way to know which side you're on is to run the division in both directions.

POSTER 7 OF 8 · PROBABILITY OUT, UNITS BACK

Fair units $d = 1/p$; implied probability $p = 1/d$ — the same division read in either direction. The CDF reads backwards too: the 90th percentile of tackles + assists is $\mu + \sigma\Phi^{-1}(0.90) = 8.6 + 2.6(1.2816) = 11.93$.